

Tan Xin Er

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience developing web, mobile, and IoT applications. Experienced in building AI-powered applications, automation tools, and intelligent systems. Passionate about full-stack development, scalable systems, and leveraging artificial intelligence to solve real-world problems.

EDUCATION

Bachelor of Engineering in Information & Communications Technology
majoring in Software Engineering (Honours with Merit) | May 2026

Singapore Institute of Technology (SIT), Singapore

Diploma in Accountancy | April 2022
Singapore Polytechnic (SP), Singapore

WORK EXPERIENCE

IT Application Intern

SBS Transit Pte Ltd

Singapore

May 2025 – April 2026

- Built an AI documentation tool using static analysis and automated diagramming to instantly map .NET and SQL architecture, logic flows, and data interactions.
- Designed and implemented backend features for an application using ASP.NET (C#). Created and maintained SQL stored procedures for new integrations, features, and workflow optimizations, ensuring strict data integrity.

Data Engineer Intern

Munich Re

Singapore

December 2024 – April 2025

- Cut data processing time by over 50% by developing automated R-based ETL workflows, streamlining data integration and reporting.

Junior Research Assistant Intern

Illinois Advanced Research Centre

Singapore

May 2024 – July 2024

- Engineered a web-based network monitoring dashboard to analyze cybersecurity risks across a complex IoT and ICS ecosystem, integrating data from robotics, microgrids, and building management systems.

PROJECTS

Secure Backend System with AWS IAM Authentication

- Executed and tested a RESTful API to support backend data retrieval and business workflows, implementing SQL-based data handling and validation logic.
- Secured API endpoints using AWS IAM (users, roles, and policies) and built a C# console application to consume and validate authenticated endpoints end-to-end.

Real-Time Multi-Modal Assistive Intelligence System

- Developed a multi-modal AI system for real-time sign language translation, speech-to-text, and facial emotion recognition using Raspberry Pi edge devices, integrating computer vision and speech recognition models with a React-based dashboard for real-time interaction.
- Engineered low-latency MQTT and Socket.IO architecture streaming sensor and AI data from edge devices to backends for real-time processing and visualization.

SKILLS

JavaScript, TypeScript, React, Node.js, REST APIs, SQL, AWS, Git, Python, C#, Flutter

AWARDS / HACKATHONS

SBS Hackathon 2025 - 1st Runner Up

- Created an ESP8266 LED control system using MQTT for real-time remote brightness adjustment, optimizing power consumption and display visibility.

CERTIFICATIONS

- SCRUM Master
- Microsoft Office Excel & Excel Expert

LANGUAGES

English: Fluent

Chinese: Intermediate

Bahasa Indonesia: Basic